

VINAYAK KUMAR SINGH

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Summary

Full Stack AI Engineer with experience building scalable backend systems, cloud-native applications, and real-time telemetry platforms. Skilled in Python, TypeScript, Node.js, REST APIs, AWS, and containerized deployments. Experienced in developing AI-assisted workflows, time-series systems, and end-to-end software solutions in Agile environments.

Education

Vellore Institute of Technology

Master of Computer Applications

2023 – 2025

CGPA: 8.26/10

Atal Bihari Vajpayee Vishwavidyalaya

Bachelor of Computer Applications

2020 – 2023

CGPA: 7.95/10

Experience

Astrome Technologies

June 2025 – Present

Full Stack AI Engineer — Python, TypeScript, Node.js, TimescaleDB, YAMCS, Docker, AWS, CI/CD Bangalore, Karnataka

- Built end-to-end payload analysis software for satellite data visualization on local maps in isolated environments.
- Developed mission operations software for satellite telemetry and telecommand using YAMCS backend infrastructure.
- Built scalable backend services using Python, Node.js, REST APIs, and TimescaleDB for analytics.
- Implemented Docker-based deployment, modular architecture, and CI/CD pipelines improving software reliability.

KIWIS & BROWNIES

Oct 2024 – Apr 2025

Software Developer — React, Vue.js, PostgreSQL, JavaScript, Git

Bangalore, Karnataka

- Developed responsive web applications using React.js and Vue.js with lazy loading optimizations.
- Optimized large dataset rendering using virtual scrolling and reactive state management techniques.
- Refactored legacy frontend modules into reusable components, improving maintainability by 30%.
- Collaborated with cross-functional teams across Germany in Agile environments for feature delivery.

Projects

Serverless Cloud Voting Platform — React.js, AWS Lambda, API Gateway, DynamoDB, IAM

May 2024

- Built serverless cloud voting platform using AWS Lambda, API Gateway, and DynamoDB.
- Developed React.js frontend integrating REST APIs for secure real-time vote processing.
- Implemented scalable event-driven architecture supporting 5,000+ daily requests with high availability.
- Configured monitoring and automated deployments achieving 99.8% uptime across application services.

Multilingual VQA Vision-Language Model — Python, Qwen2.5-VL, Unsloth, LoRA, PyTorch, RAG

March 2025

- Fine-tuned Qwen2.5-VL model for multilingual visual question answering across 14 languages.
- Built multimodal preprocessing pipelines improving training efficiency and model reasoning performance.
- Implemented parameter-efficient fine-tuning using Unsloth and LoRA for optimized training.
- Evaluated multilingual inference performance across image understanding and reasoning benchmarks.

Technical Skills

Programming Languages: Java, Python, JavaScript, C++

Frontend: TypeScript, React.js, Vue.js, HTML5, CSS3, Bootstrap

Backend & APIs: Node.js, REST APIs, Microservices Architecture

Cloud & AWS: AWS Lambda, API Gateway, DynamoDB, EC2, S3

DevOps & CI/CD: Git, Docker, CI/CD Pipelines, Postman

Databases: PostgreSQL, MySQL, DynamoDB

GenAI & AI-Assisted Development: GitHub Copilot, Claude, Cursor, Prompt Engineering

Concepts: Cloud-Native Development, Event-Driven Architecture, Agile, DevSecOps